INTERNET OF THINGS

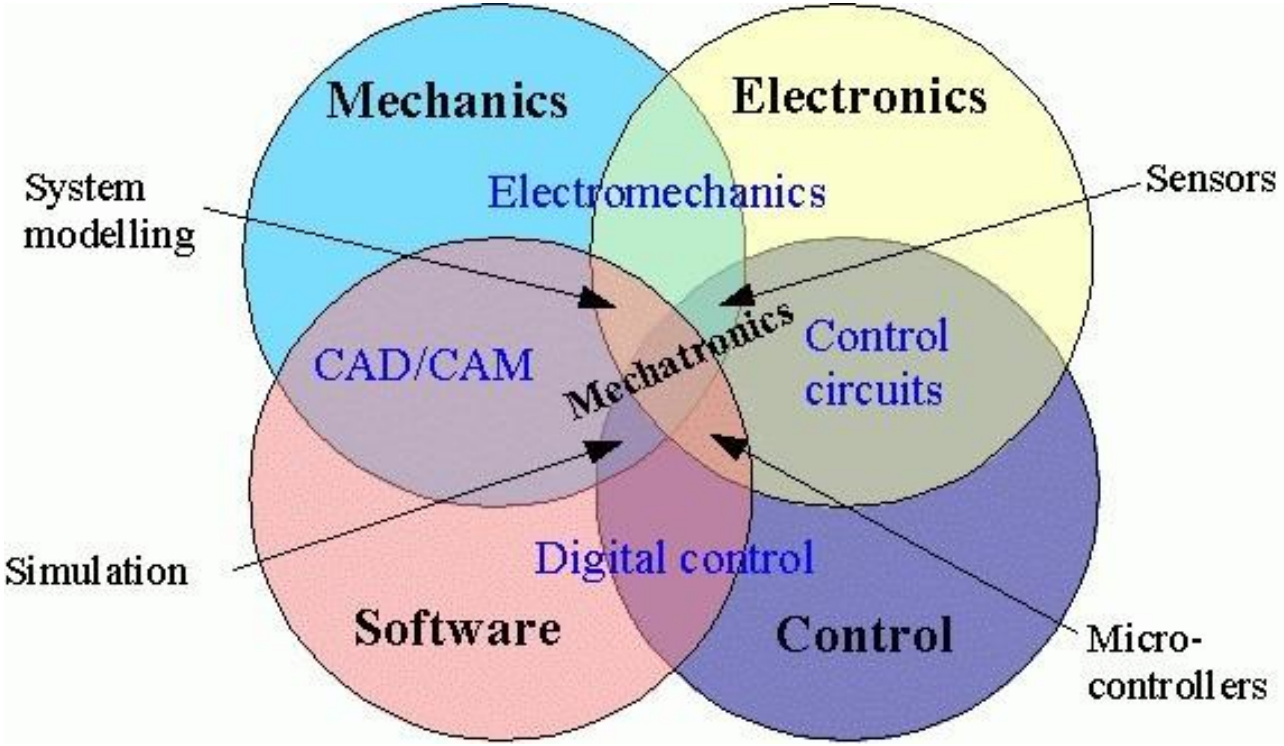
Twenty Eighth Lecture by Prof. Uday S. Dixit, Department of Mechanical Engineering, IIT Guwahati



Internet of Things

- The Internet of Things (IoT) refers to a network of physical devices, vehicles, appliances, and other items embedded with sensors, software, and network connectivity, allowing them to collect, exchange, and analyze data.
- These devices communicate with each other and with other internet-enabled devices without direct human interaction

IoT as a part of Mechatronics



https://upload.wikimedia.org/wikipedia/commons/2/2c/MechatronicsDiagram.png

Creative Commons, March15, 21

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History

- Term mechatronics was registered by an engineer named Tetsuro Mori of Yaskawa Electric Corporation in 1971.
- In 1982, Yaskawa elected to abandon its right on the word.
- Mechatronics entered the education sector in two distinct forms– as a course and as an engineering discipline.
- Microprocessor 4004 was introduced in 1971. It was using 2300 transistors and had a clock speed of 108 kHz.

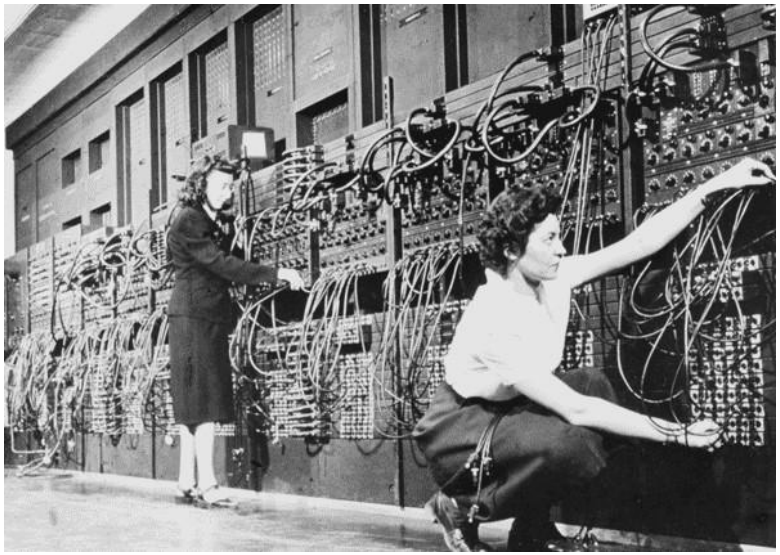
Intel Microprocessors: Historical Perspective

Processor	Year of Introduction	Number of Transistors	Initial Clock Speed	Address Bus	Data Bus	Addressable Memory
4004	1971	2,300	108 kHz	10-bit	4-bit	640 bytes
8008	1972	3,500	200 kHz	14-bit	8-bit	16 K
8080	1974	6,000	2 MHz	16-bit	8-bit	64 K
8085	1976	6,500	5 MHz	16-bit	8-bit	64 K
8086	1978	29,000	5 MHz	20-bit	16-bit	1 M
8088	1979	29,000	5 MHz	20-bit	8-bit*	1 M
80286	1982	134,000	8 MHz	24-bit	16-bit	16 M
80386	1985	275,000	16 MHz	32-bit	32-bit	4 G
80486	1989	1.2 M	25 MHz	32-bit	32-bit	4 G
Pentium	1993	3.1 M	60 MHz	32-bit	32/64-bit	4 G
Pentium Pro	1995	5.5 M	150 MHz	36-bit	32/64-bit	64 G
Pentium II	1997	8.8 M	233 MHz	36-bit	64-bit	64 G
Pentium III	1999	9.5 M	650 MHz	36-bit	64-bit	64 G
Pentium 4	2000	42 M	1.4 GHz	36-bit	64-bit	64 G

*External 8-bit and internal 16-bit data bus

From: Gaonkar, R., Microprocessor Architecture, Programming and Applications with 8085, Penram International Publishing, Mumbai

ENIAC-the first general purpose computer



In 1946, it was the first general purpose computer. It weighted 30 tons and occupied an entire room. Its computing ability can be recreated today on a silicon chip smaller than your thumbnail.

https://snl.no/ENIAC, March 15, 21 Programmering av ENIAC
Av (Ukjent)/US National Archives. ID: 594262.
Lisens: CC-PD-Mark

- The word Mechatronics is the combination of words mechanical and electronics and these two disciplines have major influence on the development of Mechatronics.
- It will be appropriate to discuss a brief history of mechanical and electronics engineering before understanding the growth of Mechatronics.

Growth of Mechatronics

- The advent of Mechatronics can be traced to the early 1950s.
- A prototype numerical control machine tool was demonstrated at Massachusetts Institute of Technology (MIT) in USA.
- In 1957, the first computer numerical control (CNC) machine was developed at MIT.
- In 1961, the part programming language Automatically Programmed Tooling (APT) was released for automatic machining of the components on CNC machines.

- In 1959, Planet Corporation introduced the first commercial robot based on limit switches and cams.
- In 1961, a Unimate robot was installed by Ford to service a die-casting machine.
- In 1978, Programmable Universal Machine for Assembly (PUMA) robot was introduced by Unimation.
- In 1979, the Selective Compliance Arm for Robotic Assembly (SCARA) was developed by Yamanshi University in Japan.

- In 1998, Honda introduced the P3 humanoid robot.
- In 1999, Sony introduced the Artificial Intelligence Robot (AIBO) robot dog.
- In 2000, Honda introduced **A**dvanced **S**tep in **I**nnovative **MO**bility (ASIMO) humanoid robot.
- In 2007, TOMY, USA introduced the i-SOBOT humanoid robot.
- By 1990, Mechatronics had established itself as an important field of engineering and *Mechatronics* journal was launched in the U.K.

Some definitions

- Mechatronics systems are the systems dealing with the movement of masses, electrons and information.
- Mechatronics is the ability to integrate electronics and computing technologies into a wide range of primarily mechanical products and processes. (*building vib. control?)*
- Mechatronics is the synergistic integration of mechanical engineering with electronics and intelligent computer control in the design of products and processes.
- Mechatronics is the application of methodology, techniques and understanding of one or more disciplines to another discipline.

- Application of complex decision making to the operation of physical systems.
- The synergetic integration of physical systems with information technology (IT) and complex-decision making in the design, manufacture and operation of industrial products and processes.
- Mechatronics is the synergetic integration of Mechanical Engineering with Electrical Engineering and/or Electronics and possibly with other disciplines too for the purpose of design, manufacture, operation and maintenance of a product.

In the late 1980s, the European Union-sponsored Industrial Research and Development Advisory Committee (IRDAC) working group came up with following definition:

“Mechatronics is the synergetic combination of precision engineering, electronic control technology and systems thinking in the design of products and processes”

Examples of Mechatronic Products

In manufacturing,

- CNC machines
- Robots
- Material handling systems
- Flexible Manufacturing Systems (FMS)

In automobiles,

- Antilock mechanism of four- wheelers
- Electronic ignition system
- Engine control system

In consumer products,

- Cameras
- Washing machines
- Toasters
- Photocopiers

In civil engineering,

- Smart structures

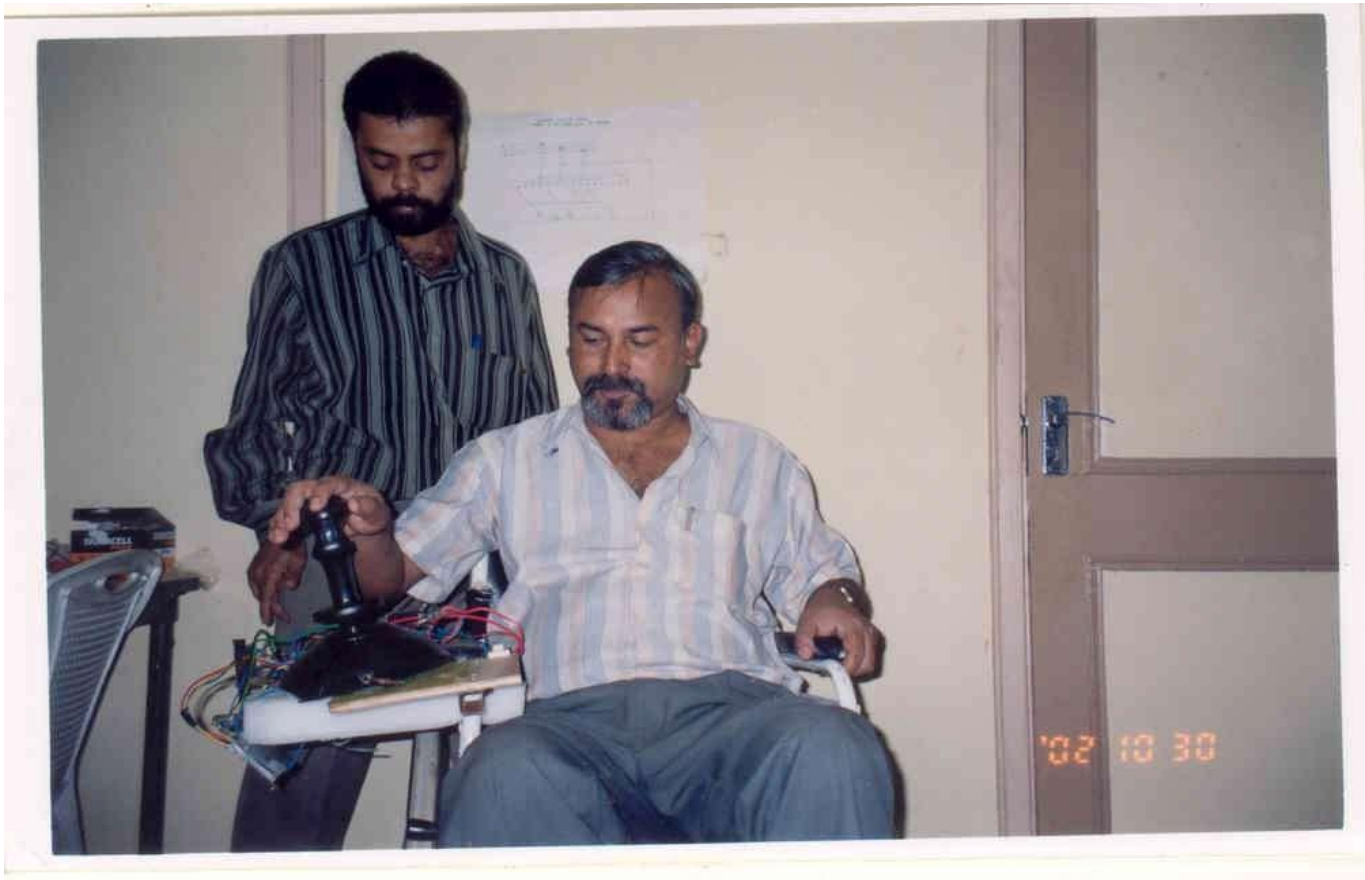
Some examples of Mechatronics projects

Projects are very important part of Mechatronics education. Project helps the student in understanding design, manufacturing, purchasing and testing aspects of Mechatronics.

Mechatronic Wheel Chair



Mechatronic Wheel Chair



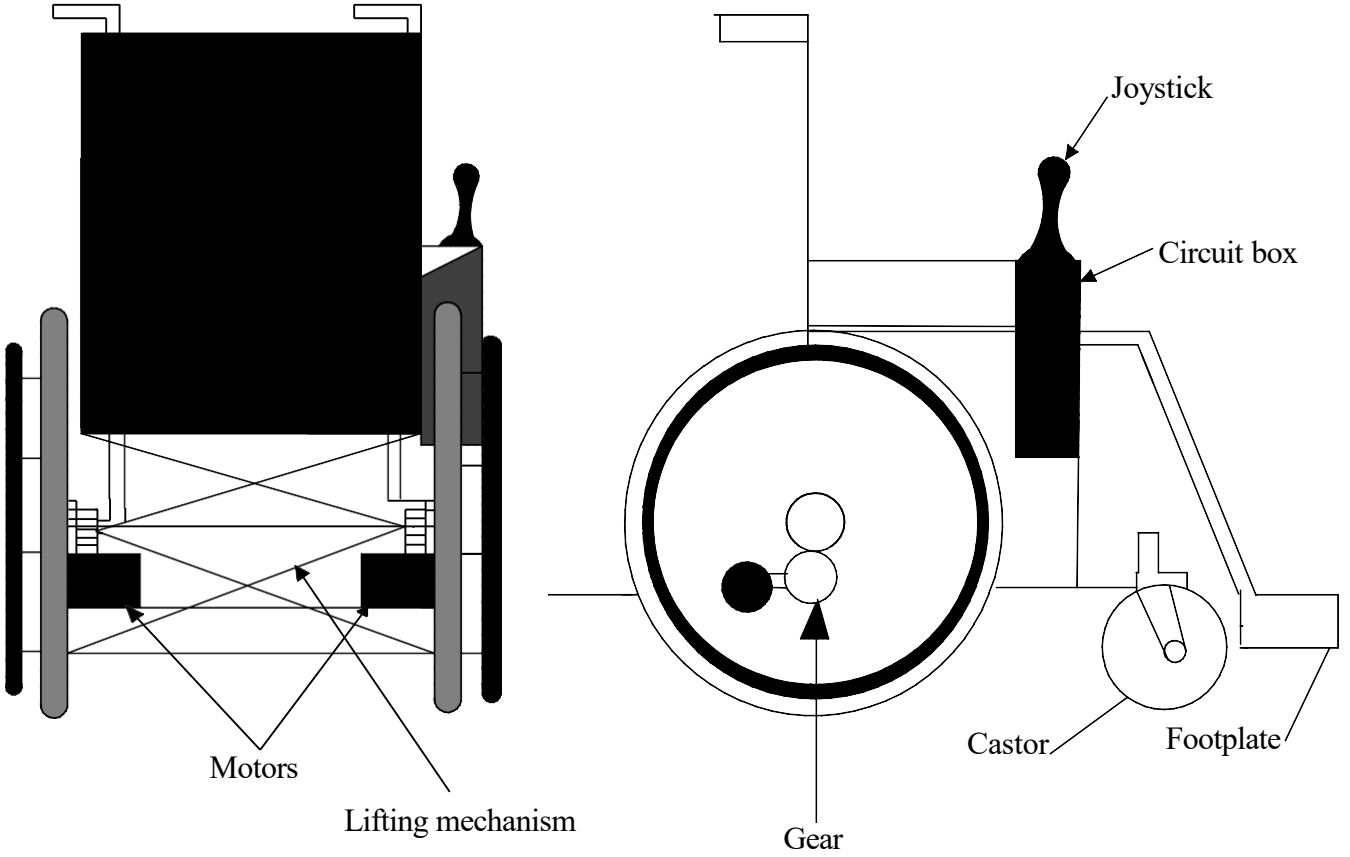


Front view

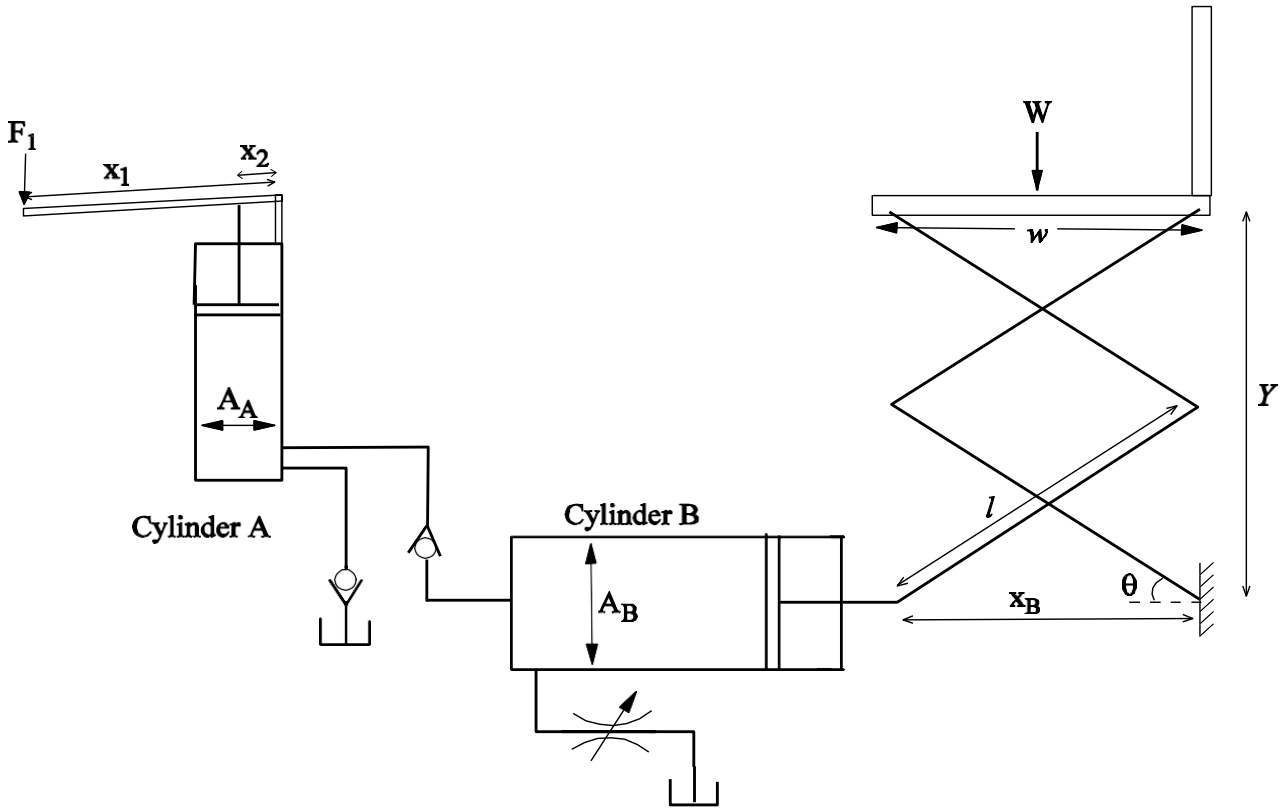
Right motor

(Permanent magnet DC geared)


'02 10 30
Battery


Joystick
Circuit box
Motors
Lifting mechanism
Gear
Castor
Footplate

Hydraulic system



Automatic gear changing system for cars

- In this project, an automatic gear changing system for cars has been developed.
- The feedback of engine speed and accelerator pedal position is taken.
- Based on a fuzzy logic program, a micro controller actuates two motors in order to bring the gear-shifting lever in a particular position.
- During gear shifting operation, the clutch paddle gets automatically disengaged.

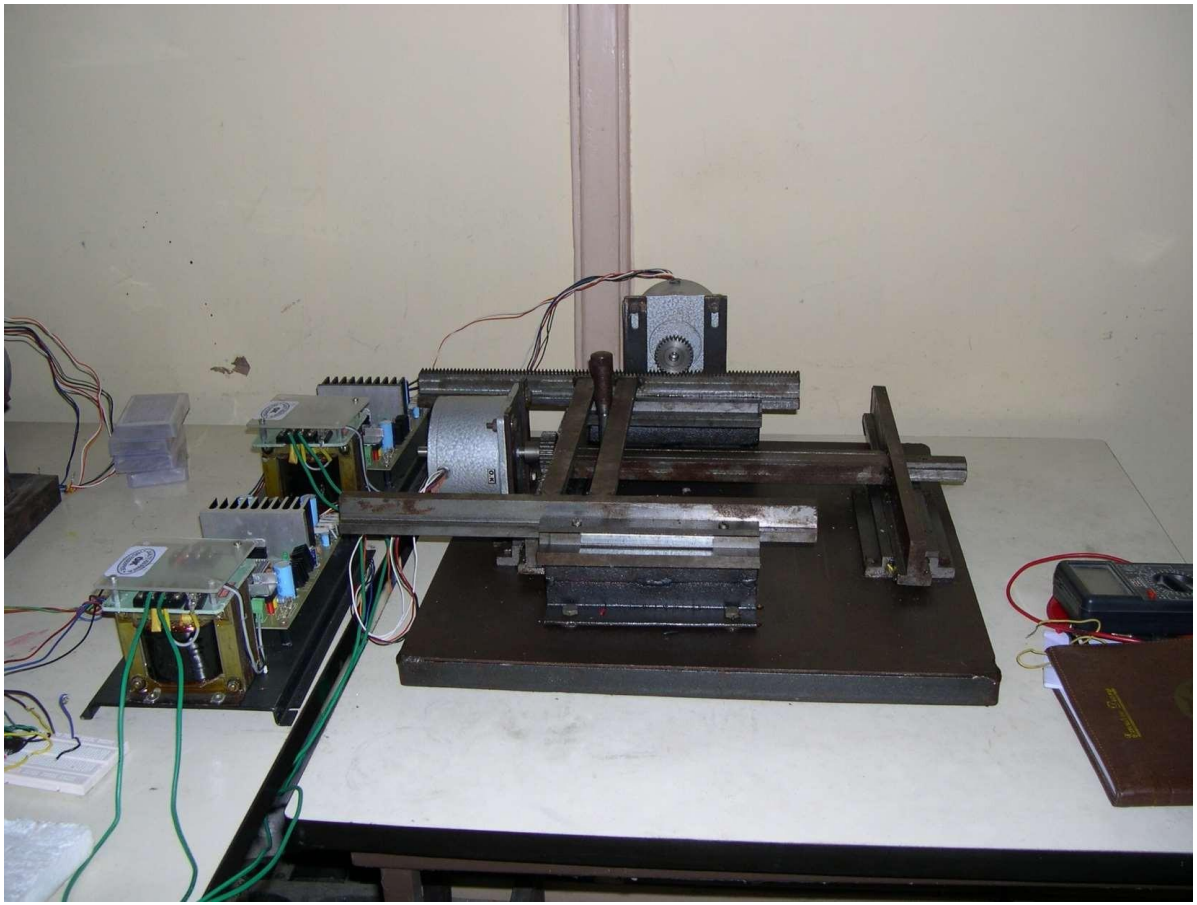


Figure. Prototype of gear change module

Low cost shop floor inspector robot

- A PC controlled shop floor inspector robot has been developed, which is driven by a (4-axis + gripper) stepper motor driven robot.
- It has a digital vernier caliper, which can be used for measuring the dimension of machined components. The inspection data can be stored in the PC.
- The cost of manufacturing of this robot is around Rs. 50,000/- (in 2001). This robot can also be used as an educational robot.

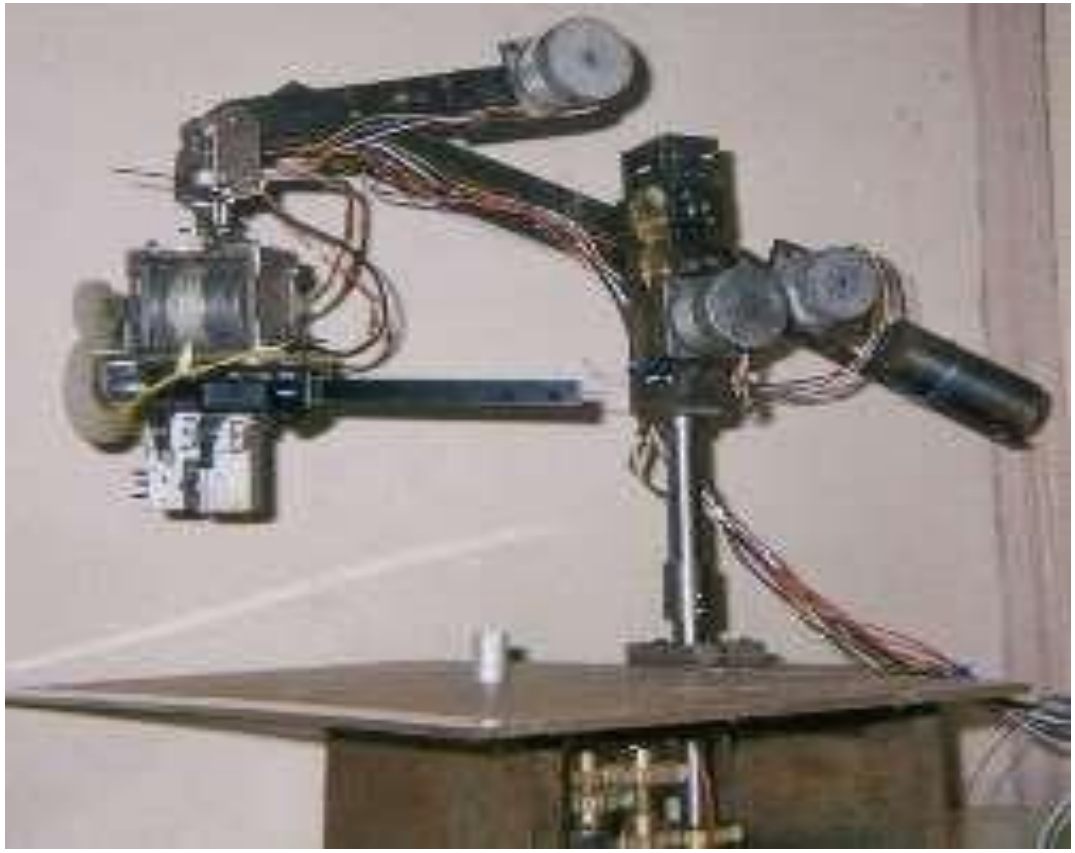
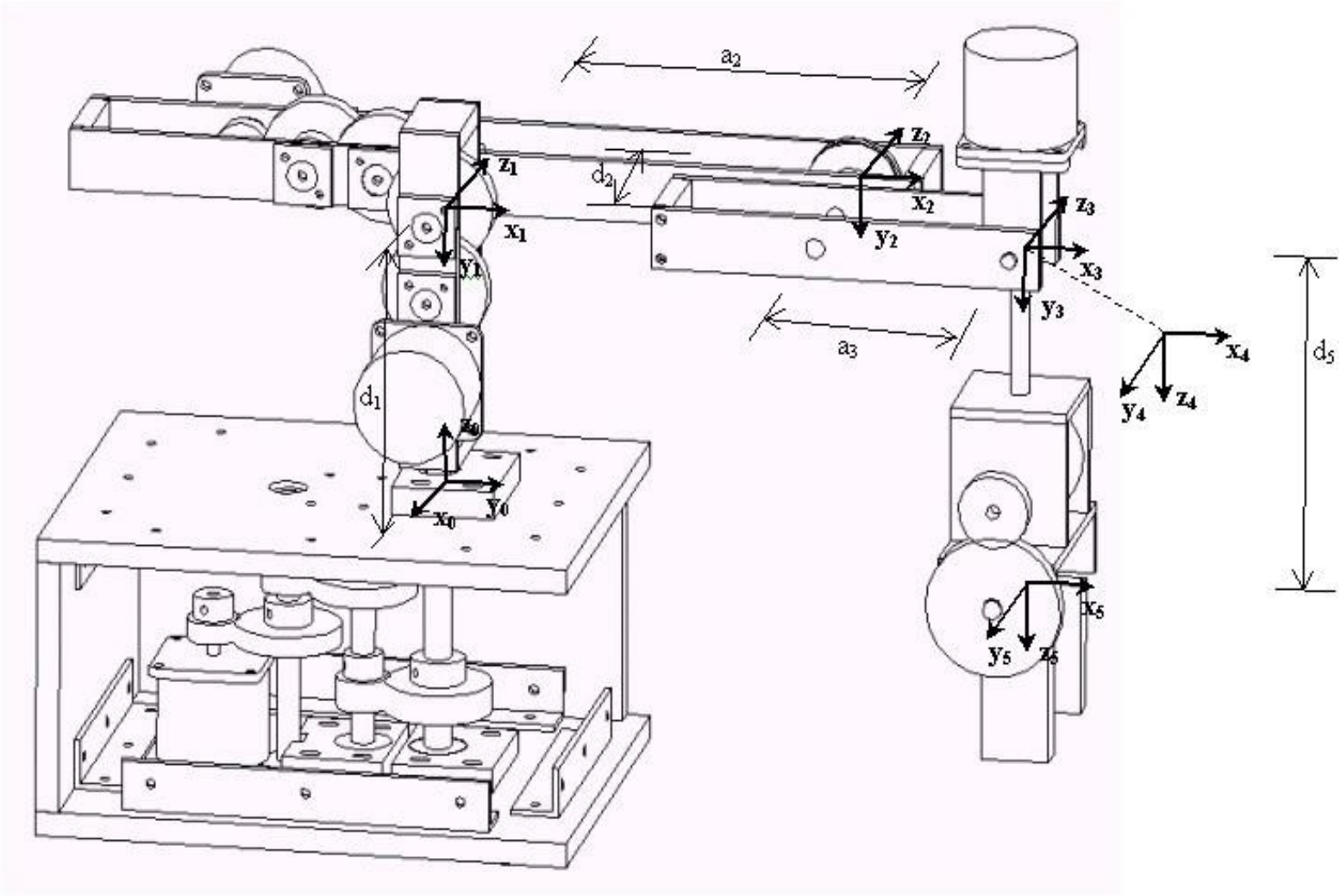
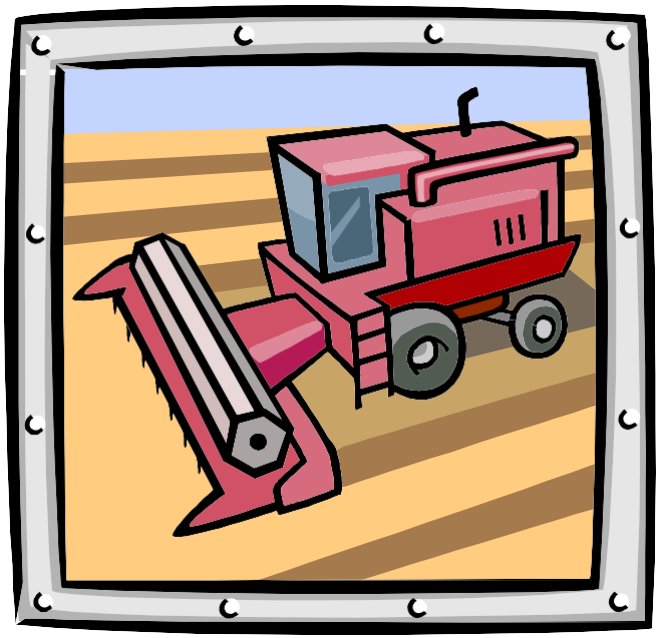


Figure. Shop floor inspector robot developed in a B.Tech. Project


a2
z2
d2
x2
y2
z3
x3
y3
z1
x1
y1
a3
x4
y4
z4
d5
d1
z0
x0
y0
x5
y5
z5

- Design is one of the oldest activities in human civilization.
- Today, a designer has choice of choosing a number of standard mechanisms and components.
- If a judicious selection of the design solution is made, an excellent design can emerge



Sensors and transducers

- Passive sensor: has no power supply and the energy for signal conditioning is drawn from the measurand.
- Active sensor: is a modulator, so it delivers more energy to next stage than it draws from the measurand

Limit switch



Mechanical limit switch

- Consists of an actuator arm that operates a set of electrical contacts when displaced
- Actuator is **lever** type and **plunger** type

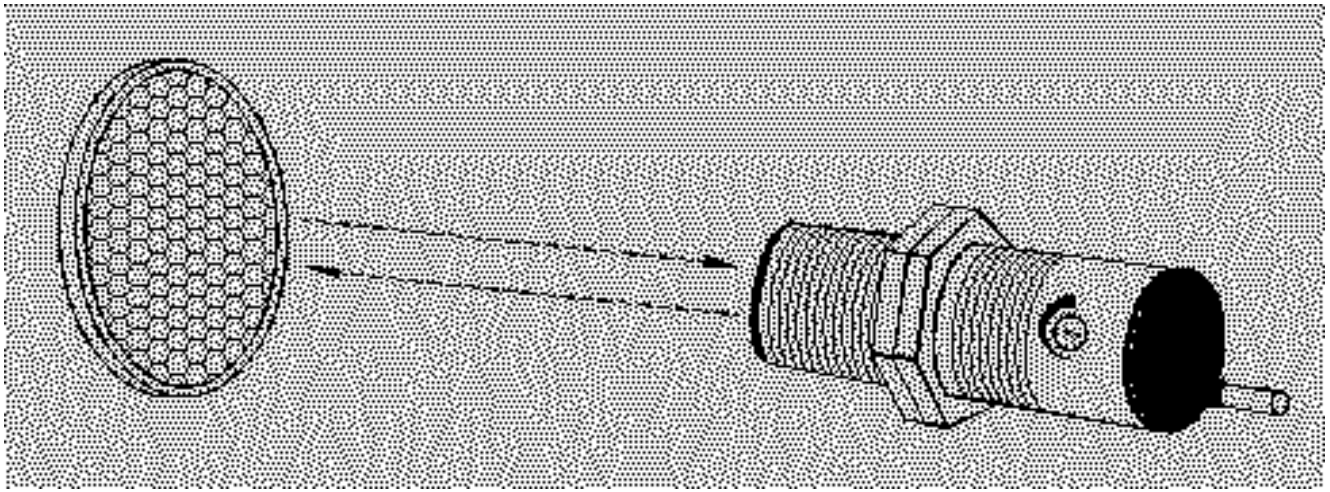
Proximity sensor



Proximity sensors

- Non-contact sensor that works by electromagnetic induction
- Inductive proximity sensor-can detect ferrous or non-ferrous metallic object.
- Capacitive proximity sensor-can detect non-metallic objects as well.

Retro-reflective photoelectric sensor



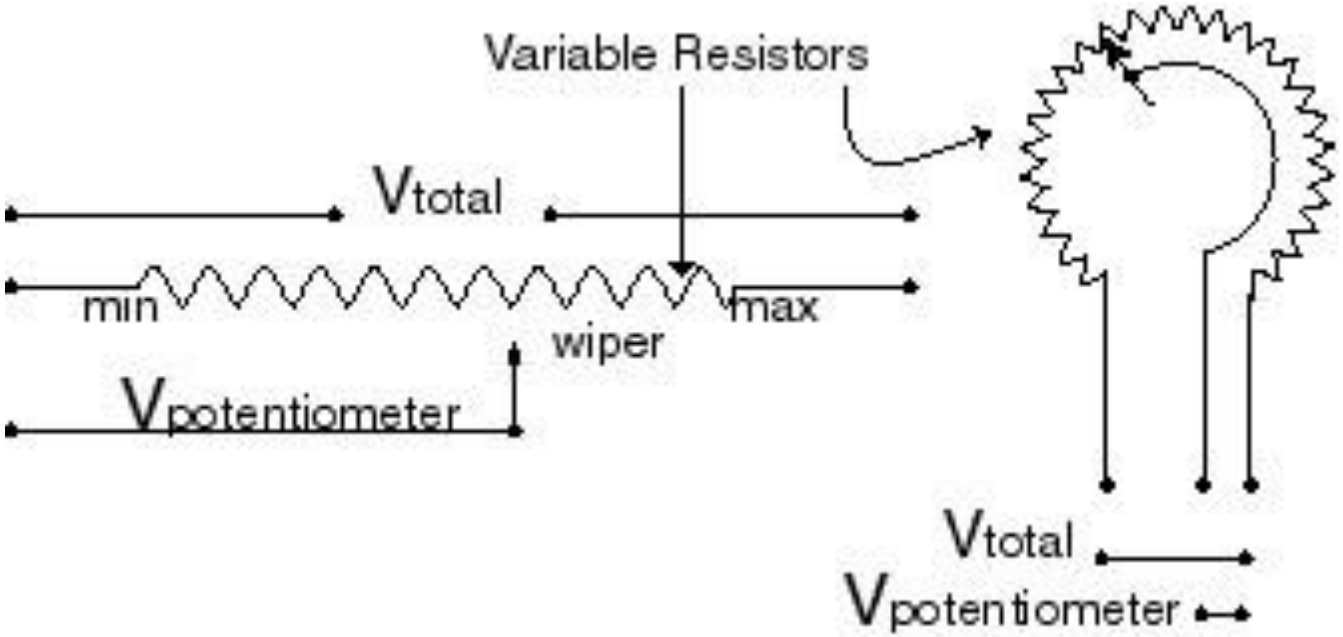
Fluid flow switch

- Discrete event sensor
- Used to detect overflow of liquid in a pipe
- Can be used to govern the speed of an upstream pump

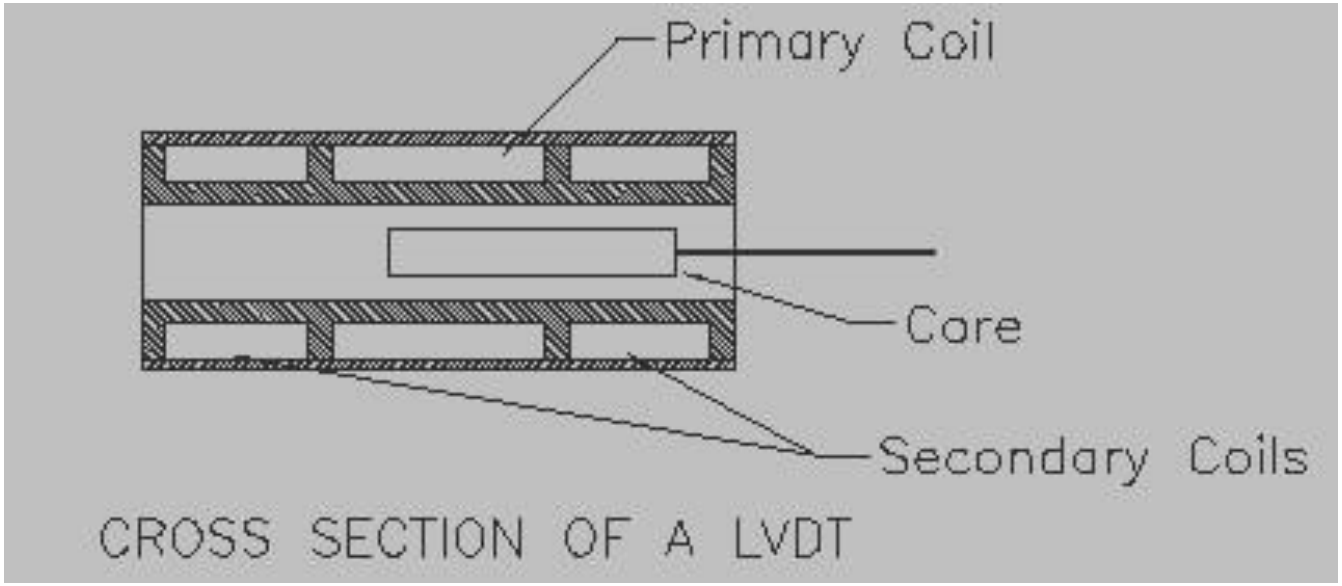
Reed Switch

- Contact-less input device
- Contacts changeover in the proximity of a magnet and on removing it, they change back
- Easy to mount and unaffected by foreign particles
- Affected by external magnetic fields

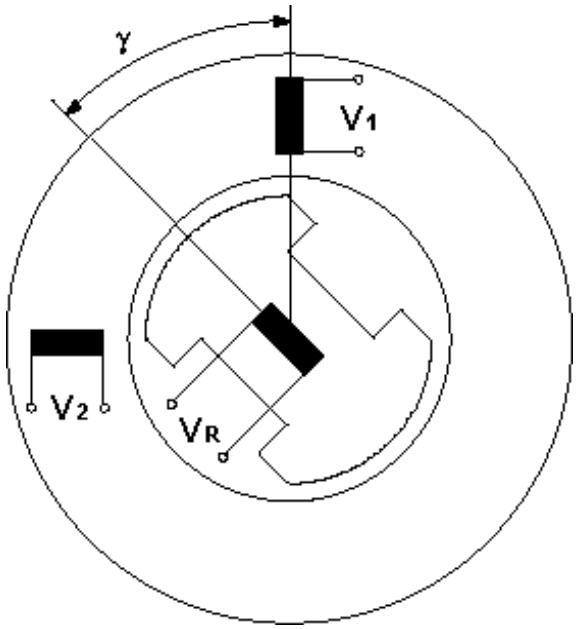
Potentiometers



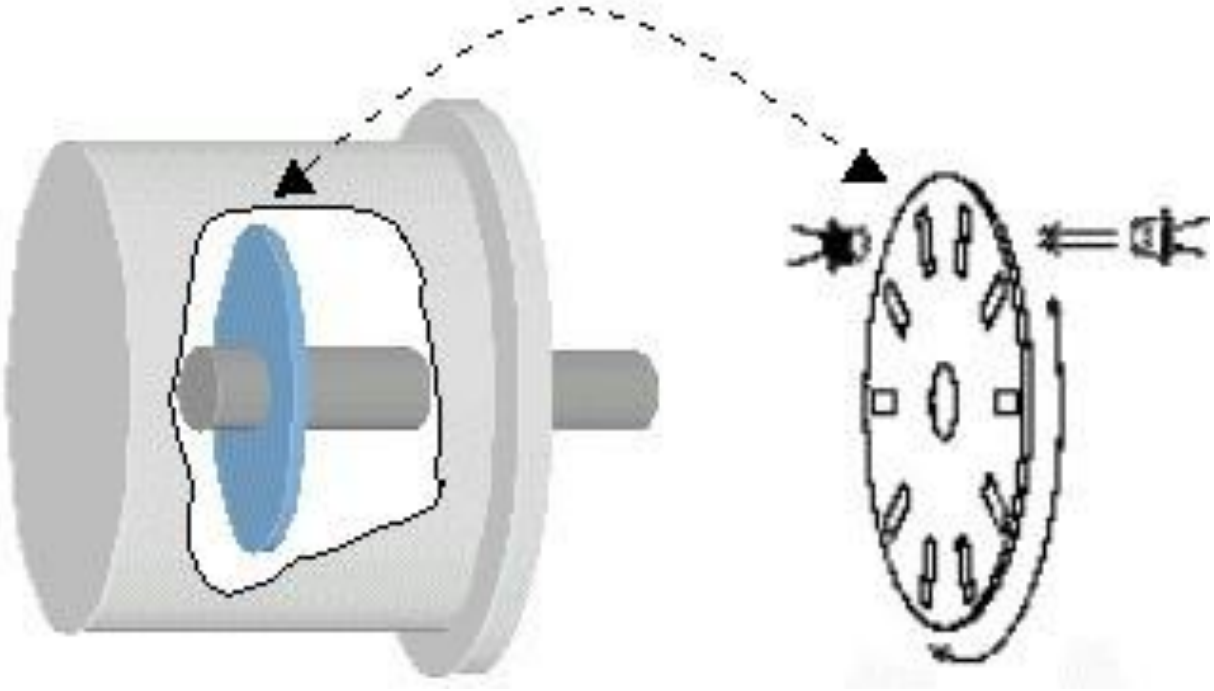
LVDT



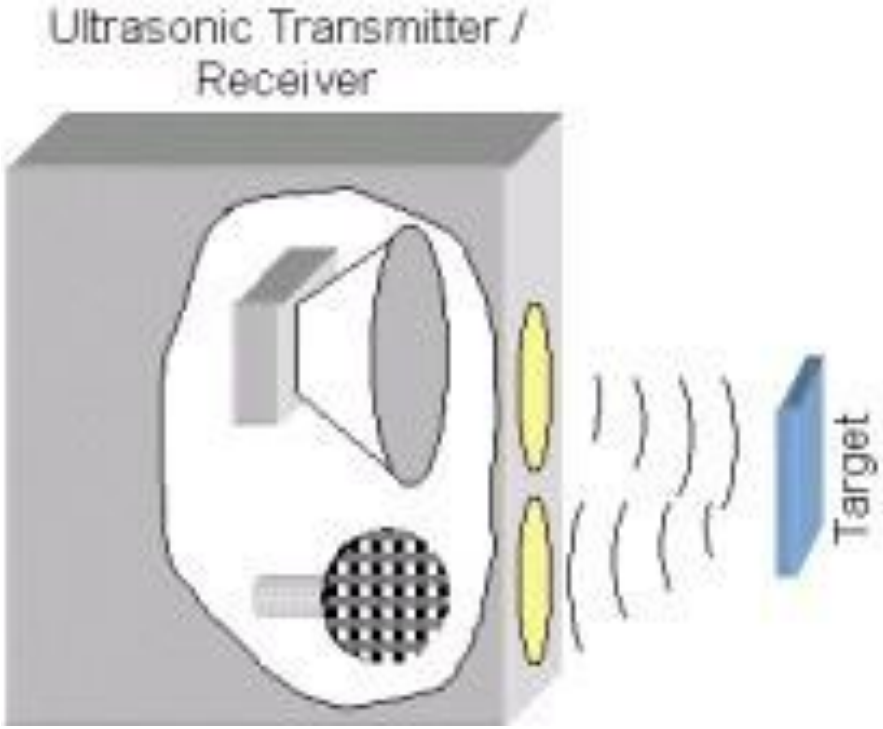
Resolver



Optical encoders



Ultrasonic range sensors



Tachometers

- Analog velocity measuring device
- Tachometers operate on the principle of a DC generator or an AC generator.

Smart Sensors

Purpose of development:

- Compensation for non-ideal behavior of sensors
- Provision for communication of process data with the ‘host’ system

Hardware Technologies in IoT

- **Arduino** – Open-source microcontroller board; ideal for prototyping.
- **Raspberry Pi** – Low-cost, full-fledged Linux computer; good for edge computing.
- **ESP8266 / ESP32** – Wi-Fi-enabled microcontrollers from Espressif; very popular in IoT due to low cost and built-in connectivity.
- **BeagleBone** – Embedded Linux computer for industrial applications.
- **STM32** – ARM Cortex-M-based microcontrollers used in commercial IoT applications.

Connectivity technologies

- **Wi-Fi** – Suitable for short-range, high-data-rate communication.
- **Bluetooth / BLE** – Used in wearables and short-range low-power applications.
- **Zigbee / Z-Wave** – Low-power mesh networking protocols used in smart homes.
- **LoRa / LoRaWAN** – Long-range, low-power wireless communication for rural and industrial areas.
- **NB-IoT / LTE-M** – Cellular technologies designed specifically for IoT.
- **Ethernet** – Reliable wired connectivity, especially in industrial setups.
- **5G** – High-speed, low-latency connectivity for massive IoT.

Cloud Platform & Services

AWS IoT Core

Microsoft Azure IoT Hub

Google Cloud IoT

IBM Watson IoT

ThingSpeak – Popular in academia and prototyping.

Blynk – Mobile app platform for Arduino and similar boards.

Software and operating systems

FreeRTOS – Real-time operating system for embedded devices.

Mbed OS – Lightweight OS for IoT devices from ARM.

Raspberry Pi OS (Linux-based) – Used in Raspberry Pi boards.

Arduino IDE – Popular environment for coding Arduino devices.

PlatformIO – Cross-platform IDE for embedded development.

Node-RED – Visual programming tool for wiring IoT systems.

Security Technologies

- **TLS/SSL** – Secure data transfer protocols.
- **MQTT with TLS** – Lightweight messaging protocol with encryption.
- **Device Authentication Tools** – Such as TPM chips or digital certificates.
- **Firmware Over-The-Air (FOTA)** – Secure firmware updates.

Data Analytics & AI

For making sense of IoT data.
Edge AI frameworks – e.g., TensorFlow Lite, TinyML.
Cloud-based analytics – AWS, Azure, or GCP offer ML tools.
Time-series databases – InfluxDB, Prometheus.